Course Name: Database Systems	**Session:** 2025
Lab/Week: 1	**Semester:** Spring 2026
Lab Title: Installation and Configuration of MySQL	**Date:** 10-2-2026

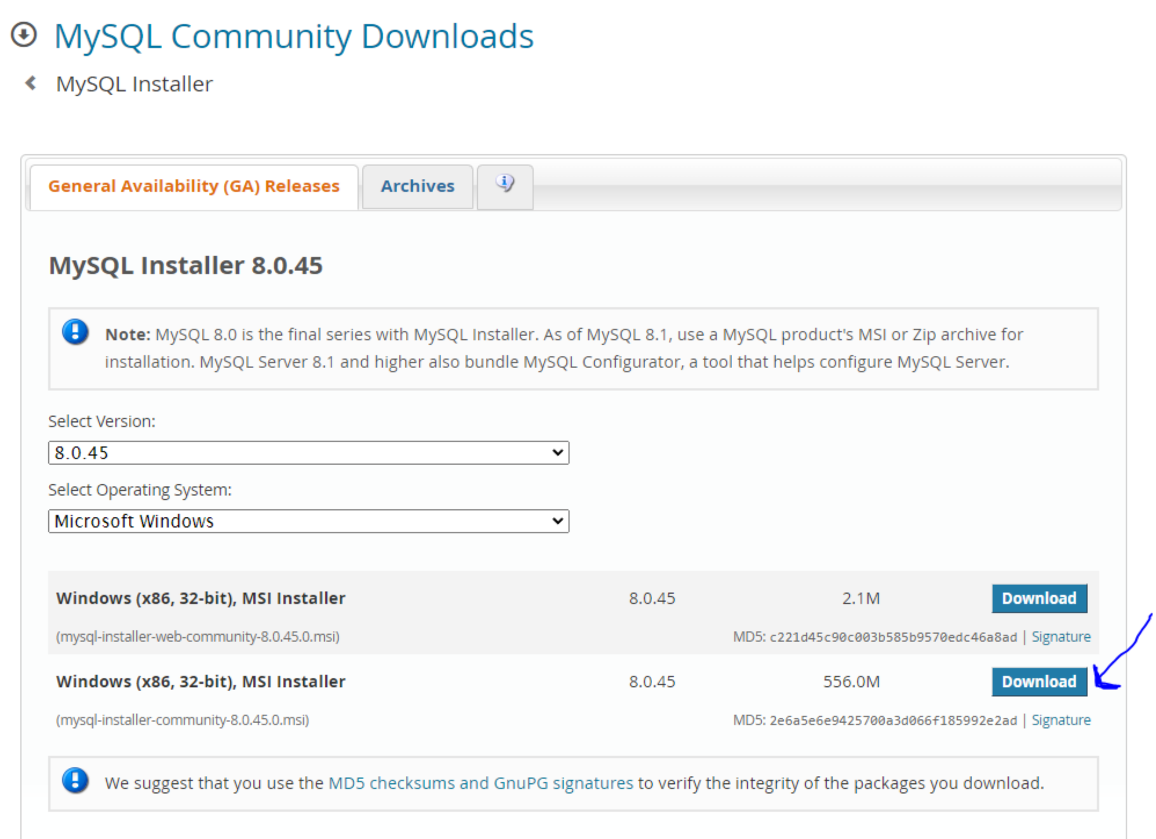
Objectives:

- Install MySQL
- Create Database
- Import Sample Dataset
- Export Database
- Create User+Permissions
- Connect from another PC

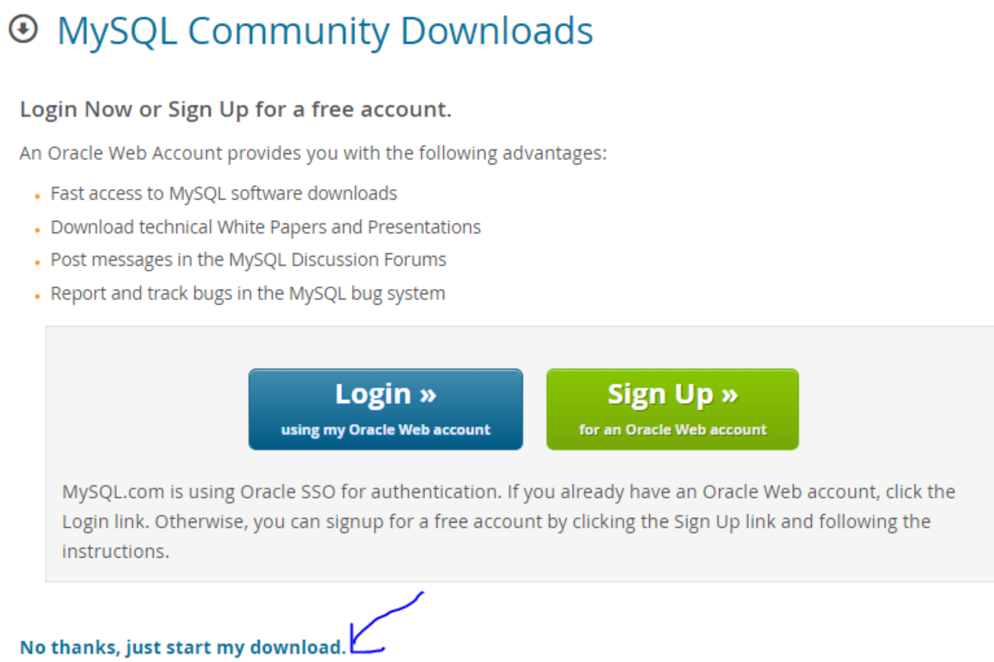
Lab Task1: Install MySQL

Step 1: Download MySQL Installer

1. Download MySQL Installer from: https://dev.mysql.com/downloads/installer/
2. Click on Windows (x86, 32-bit), MSI Installer



3. Click on No thanks, just start my download to proceed without creating an account.



Step 2: Launch the Installer

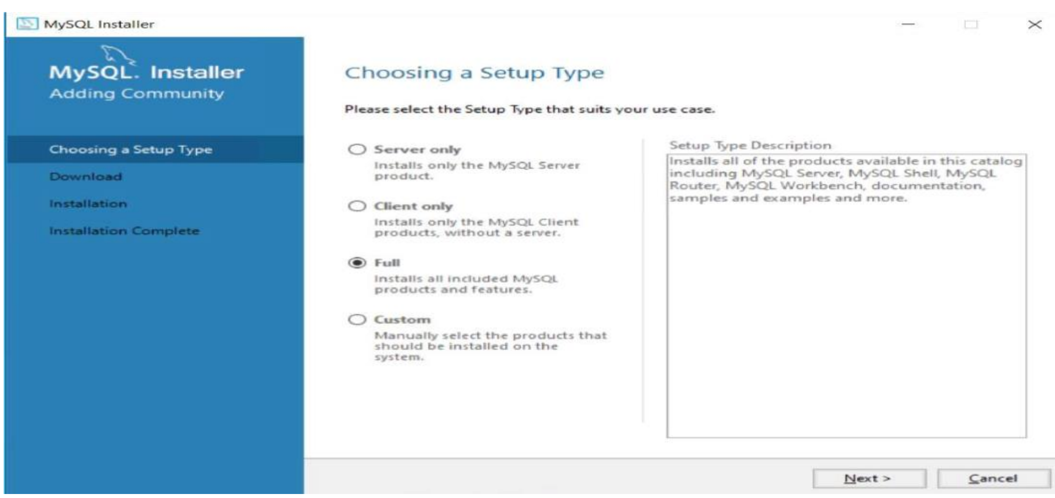
1. Navigate to your Downloads folder and double-click the downloaded file to start the installation.
2. If prompted by Windows User Account Control, click Yes to allow the installer to run.

Step 3: Select Setup Type

1. On the Setup Type screen:

- Choose Full (recommended for a complete installation with all features).

2. Click Next.

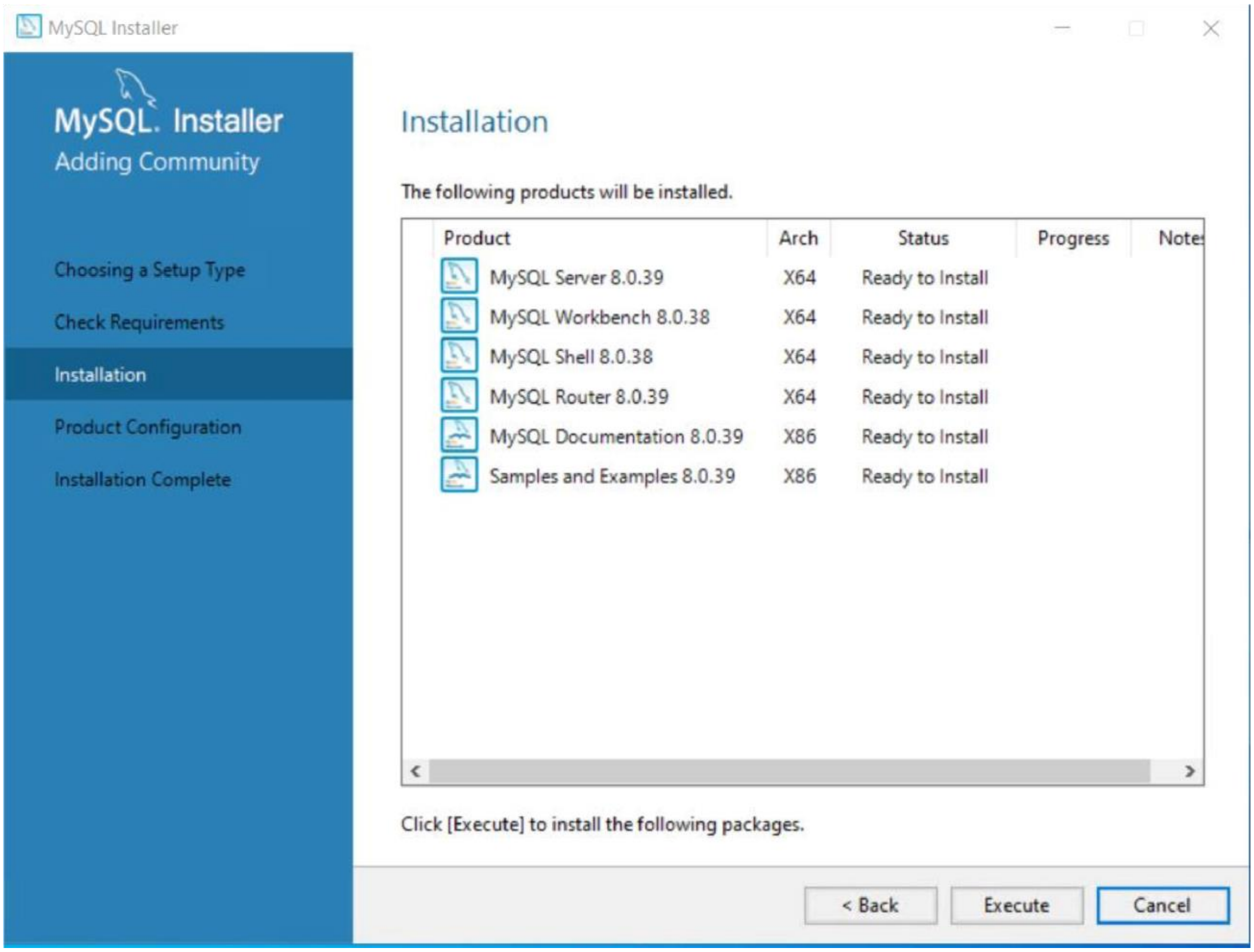


Step 4: Install Required Components

1. On the Check Requirements screen, ensure all required components are listed.

2. Click Execute to download and install the required components.

- If a license agreement pops up, click I Agree.



3. Wait for the process to complete, and then click Next.

Step 5: Configuration

1. On the Product Configuration screen, click Next.
2. Type and Networking:

- Leave the default settings (e.g., Development Computer).
- Click Next.

Step 6: Authentication Method

1. On the Authentication Method screen:

- Select "Use Strong Password Encryption for Authentication (Recommended)".

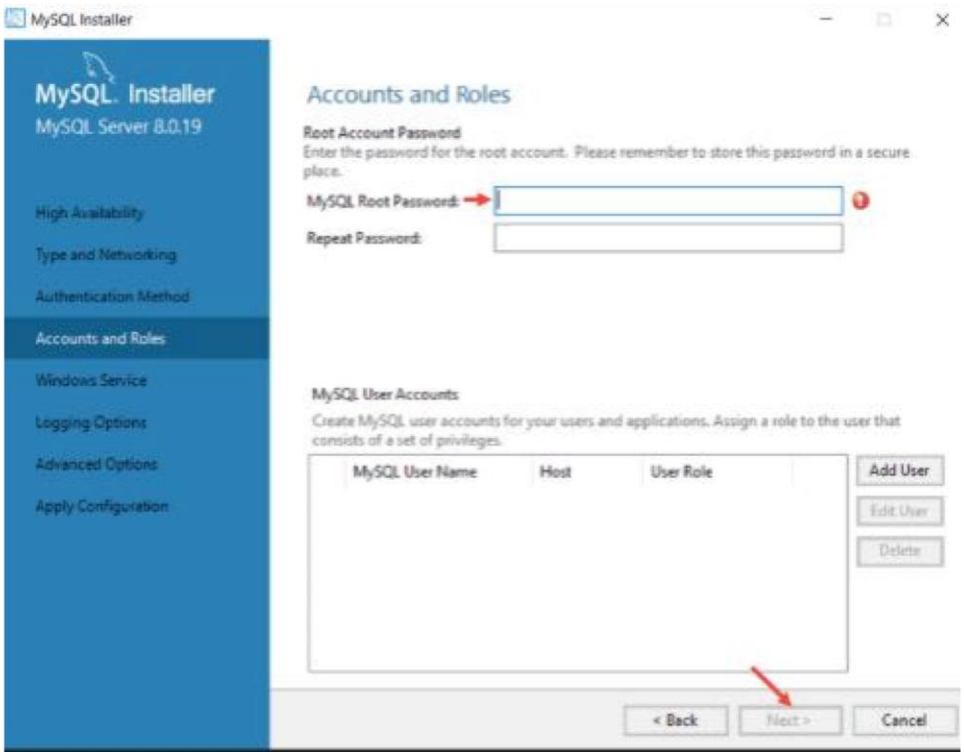
2. Click Next.

Step 7: Set Root Password

1. On the Accounts and Roles screen:

- Enter a strong password for the root user.
- Example: Choose a mix of uppercase, lowercase, numbers, and symbols.
- **Important:** Save this password securely, as it will be required for accessing the database.

2. Click Next.



Step 8: Service Configuration

1. Leave the Windows Service Configuration settings as default.

- MySQL will automatically start as a service on system boot.

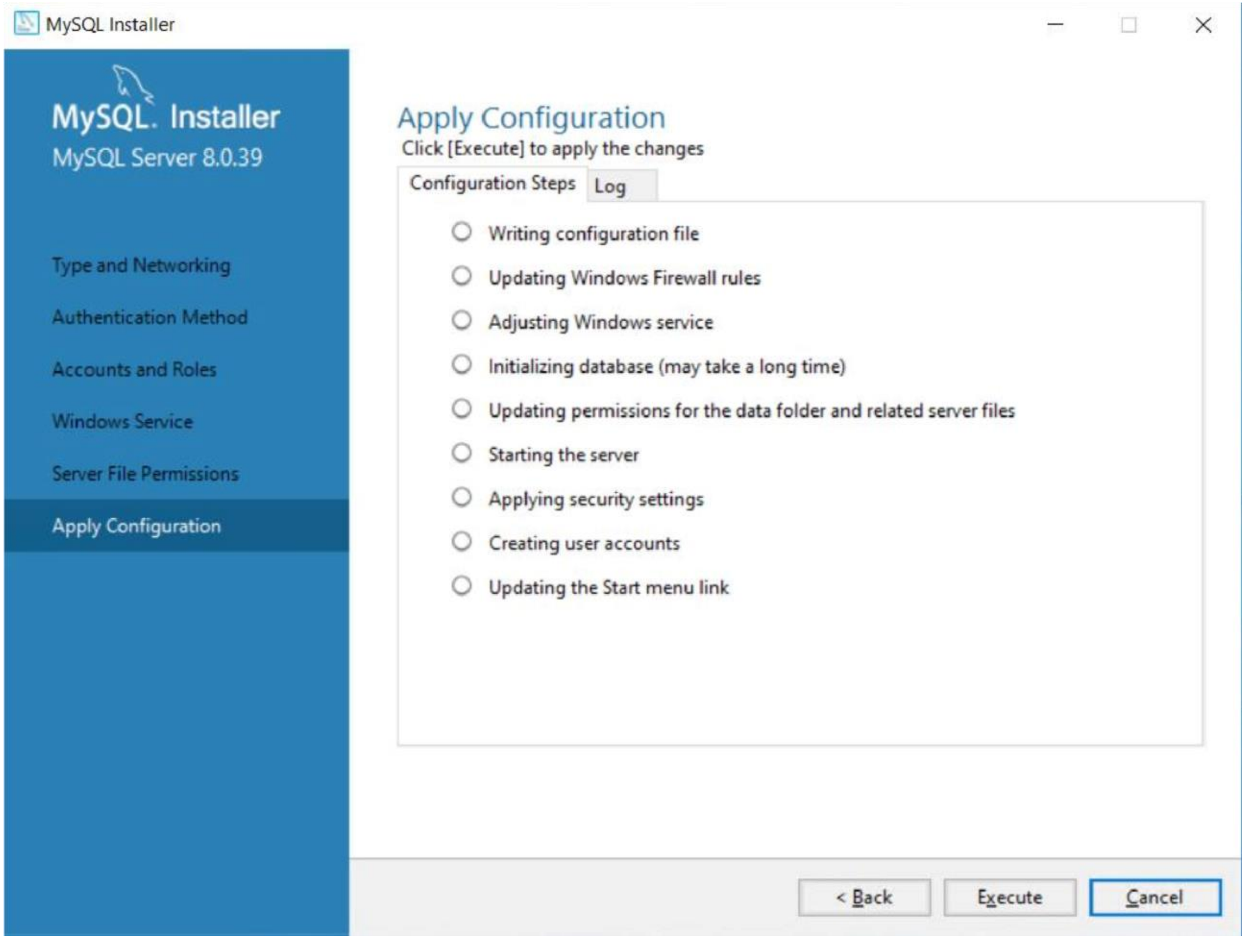
2. Click Next.

Step 9: Apply Configuration

1. On the Apply Configuration screen, click Execute to apply all settings.

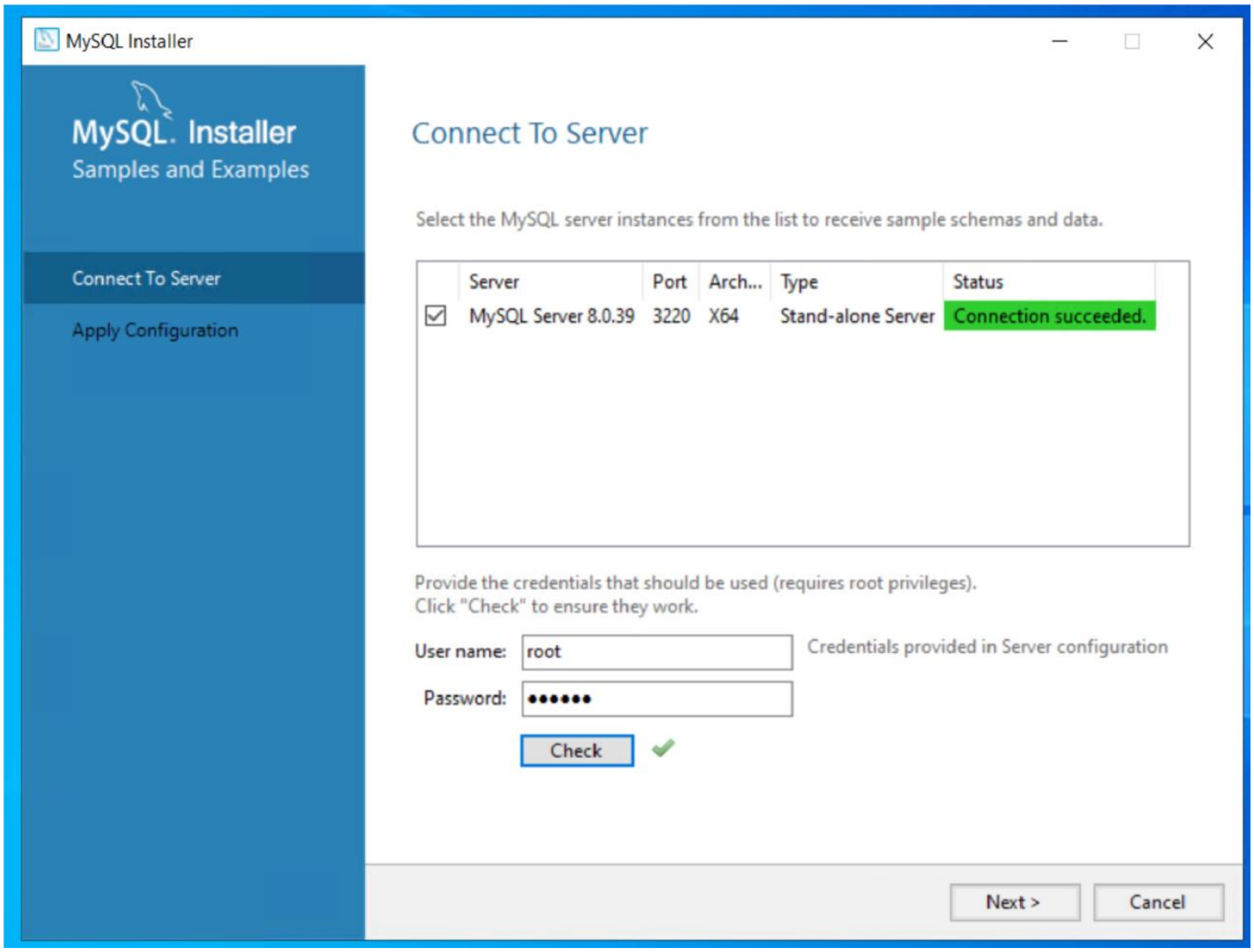
2. Wait for the configuration process to complete.

3. Once completed, click Finish.



Step 10: Test Server Connection

1. On the Connect to Server screen:
   - Enter the root username and the password you set earlier.
2. Click Check to verify the connection.
   - If successful, you’ll see a confirmation message.
3. Click Next.



Step 11: Finish Installation

1. On the final screen, click Finish to complete the installation.
2. If a MySQL console appears, you can close it.

Post-Installation Steps

1. Open MySQL Workbench or any MySQL-compatible client to connect to your server.
2. Use the credentials (root and your password) to log in and begin creating databases or executing SQL queries.

Lab Task 2: Create Database

Step 1: Open MySQL Workbench

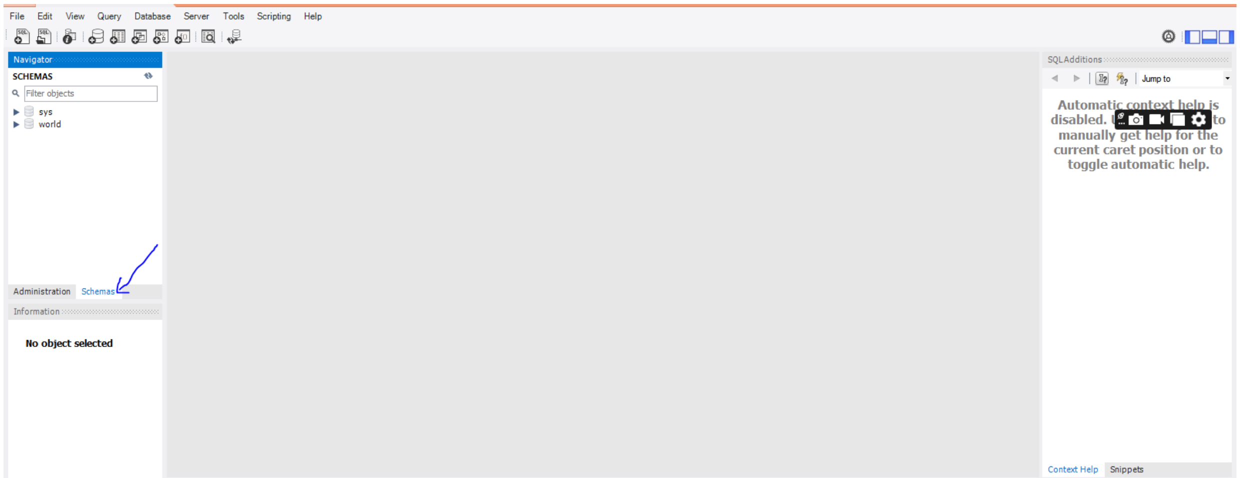
1. Launch MySQL Workbench from the Start Menu or desktop.

2. Under the "MySQL Connections" section:

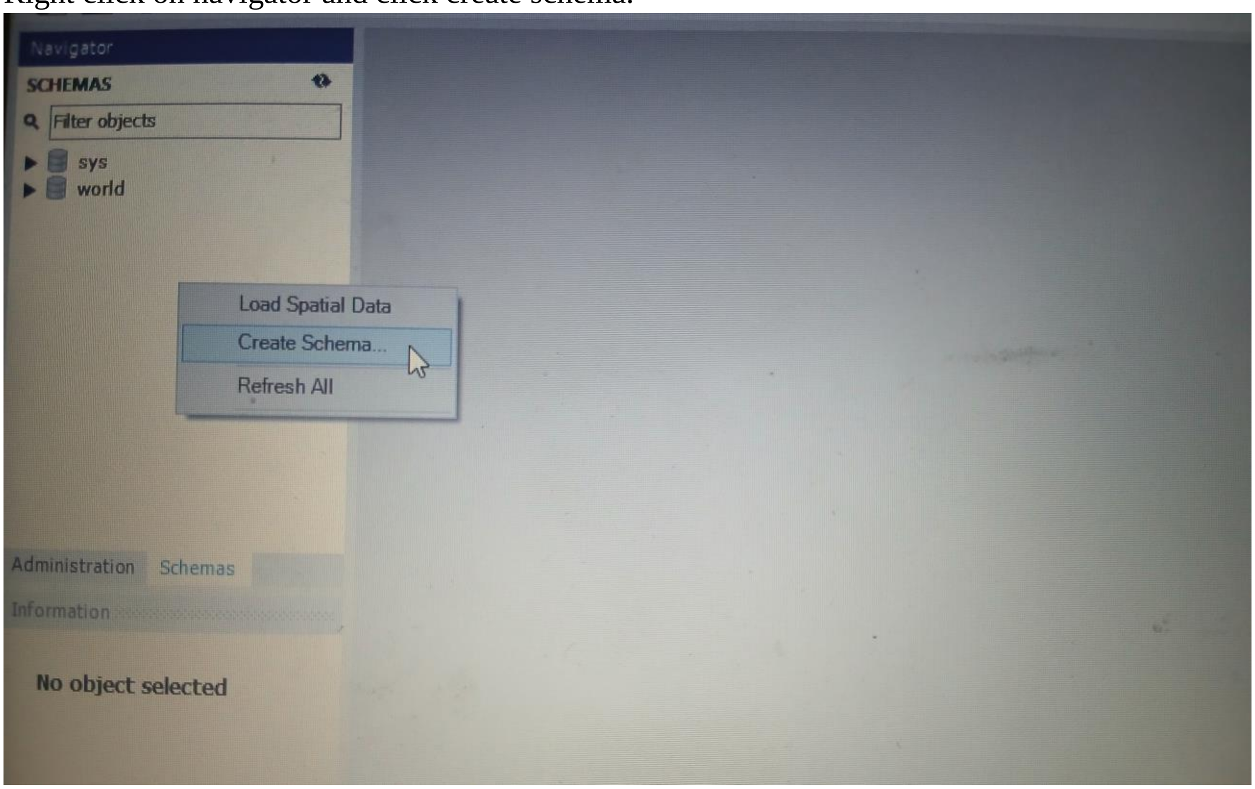
- Click on your database connection (e.g., Local Instance MySQL).
- Enter your root password (set during installation) if prompted.

Method 1:

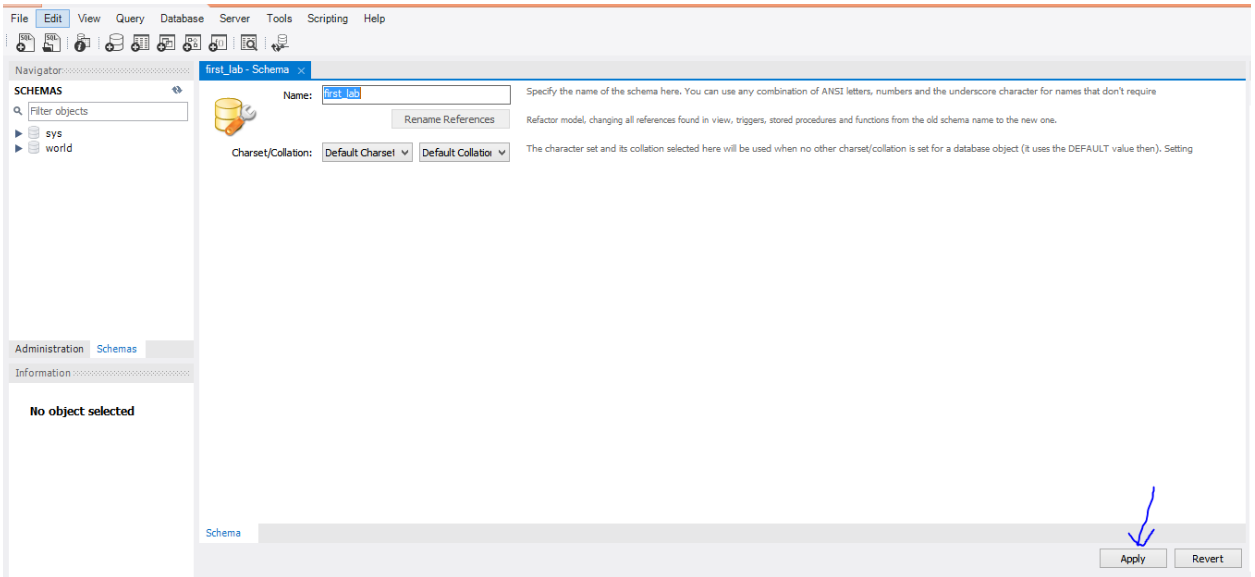
1. Go to Schemas



2. Right click on navigator and click create schema.

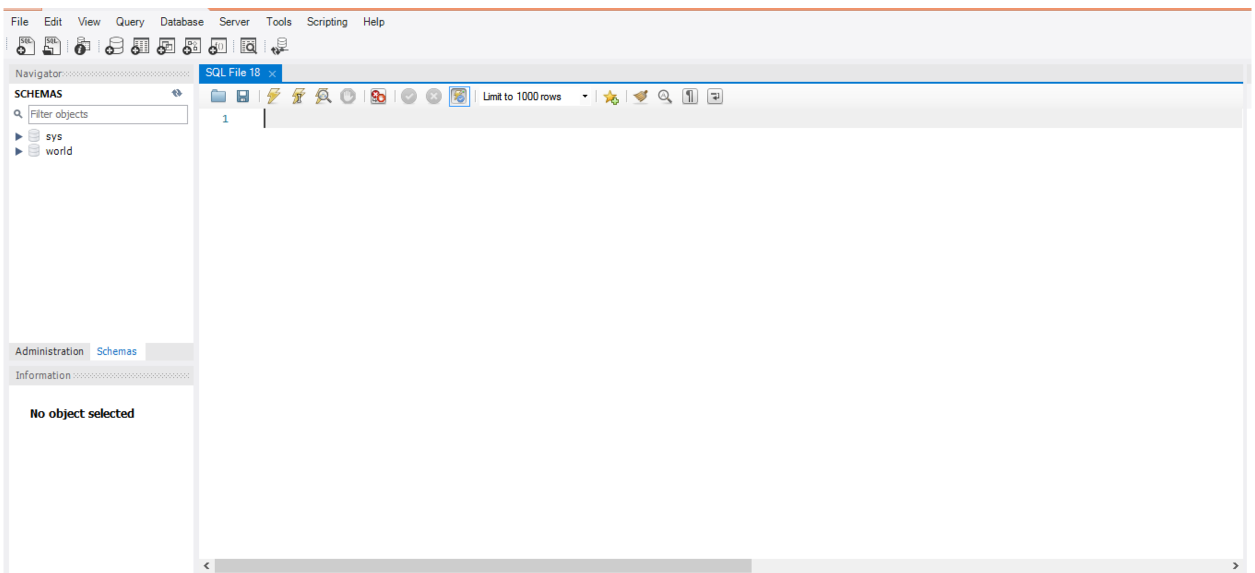


3. Write the name of database and then click Apply to create database.

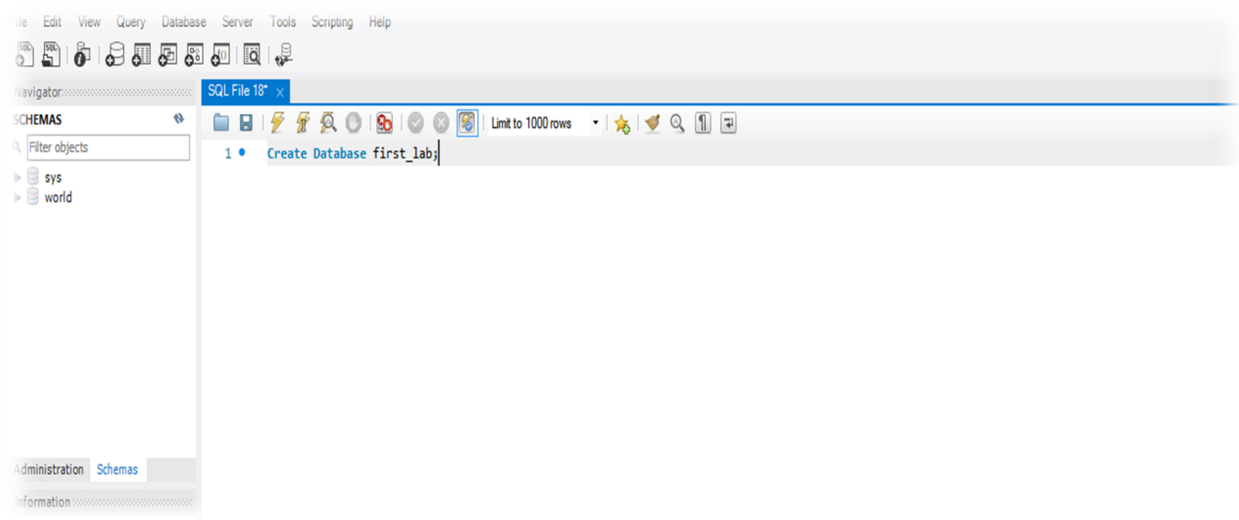


Method 2:

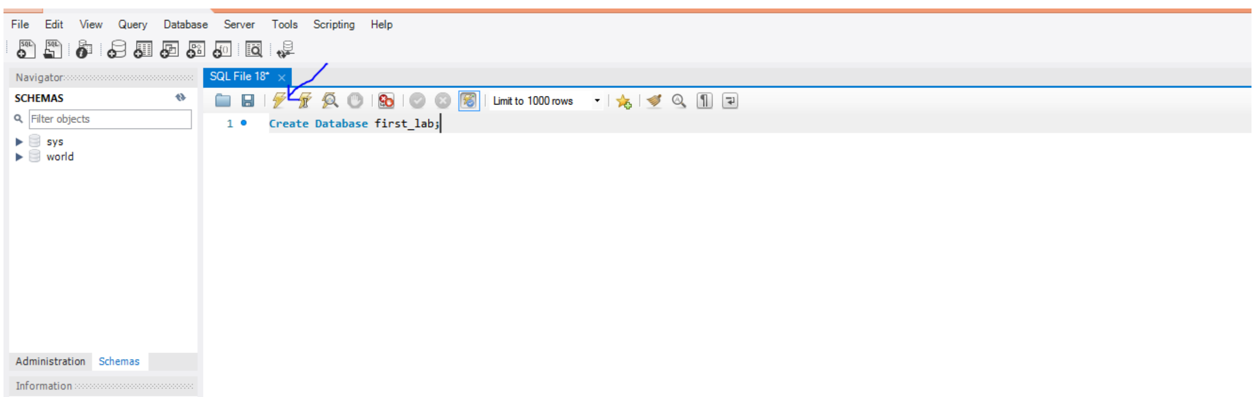
1. Navigate to the top menu and click on "File" > "New Query Tab" (or press Ctrl+T).



2. To create database, write below query on query tab.



3. Click on  to create database.



Step 3: Verify the Database Creation

1. On the left-hand Schema Browser pane, right-click and select "Refresh All".

2. Look for the first_lab database in the list of schemas.

Step 4: Set first_lab as the Active Database

1. To ensure you're working with the Northwind database, execute the following command:

   USE first_lab;

2. Click on the "Execute" button to set Northwind as the current database.

Step 5: Confirm the Database Exists

1. Run the following command to list all databases in your MySQL instance:
   SHOW DATABASES;

2. Verify that first_lab appears in the list of databases.

Lab Task 3: Restore database schema by executing to comprehend how schema scripts define database structure.

Step 1: Select the Target Database

1. In the left-hand Schemas pane:

- Right-click on the first_lab database.
- Select "Set as Default Schema" to ensure that all operations target this database.

Step 2: Open the Schema Script

1. In the top menu, click on File > Open SQL Script.

2. Navigate to the folder containing provided first_lab.sql.

3. Select the first_lab.sql file and click Open.

Step 3: Review the Schema Script

1. The contents of the first_lab.sql script will appear in the query editor.

2. Review the script to understand its structure:

- CREATE TABLE statements define the tables.

Step 4: Execute the Schema Script

1. Click on the "Execute" button (lightning bolt icon) in the toolbar or press Ctrl+Shift+Enter.

2. Monitor the Action Output pane at the bottom:

- Look for messages indicating successful execution (e.g., Table Created).

Step 5: Verify the Schema Restoration

1. Refresh the Schemas pane by right-clicking and selecting "Refresh All".

2. Expand the first_lab database to view the restored structure:

- Check the Tables, Views, and Stored Procedures folders.

3. Run the following query to list the tables in the database:

Show TABLES;

Lab Task 4: Populate the database with data to practice importing data into a structured database.

Step 1: Select the Target Database

1. In the left-hand Schemas pane:

- Right-click on the first_lab database.
- Select "Set as Default Schema" to ensure that the data is imported into the correct database.

Step 2: Open the Data Script

1. In the top menu, click on File > Open SQL Script.
2. Navigate to the folder containing the first_lab_data.sql file.
3. Select the first_lab_data.sql file and click Open.

Step 3: Review the Data Script

1. The contents of the first_lab_data.sql script will appear in the query editor.

2. Review the script to understand its structure:

- Look for INSERT INTO statements, which populate the tables with data.

Step 4: Execute the Data Script

1. Click on the "Execute" button (lightning bolt icon) in the toolbar or press Ctrl+Shift+Enter.

2. Monitor the Action Output pane at the bottom:

- Look for messages indicating successful execution (e.g., Rows Inserted).

Lab Task 5: Create a new database to practice the fundamental steps of setting up a database environment for structured data storage and management

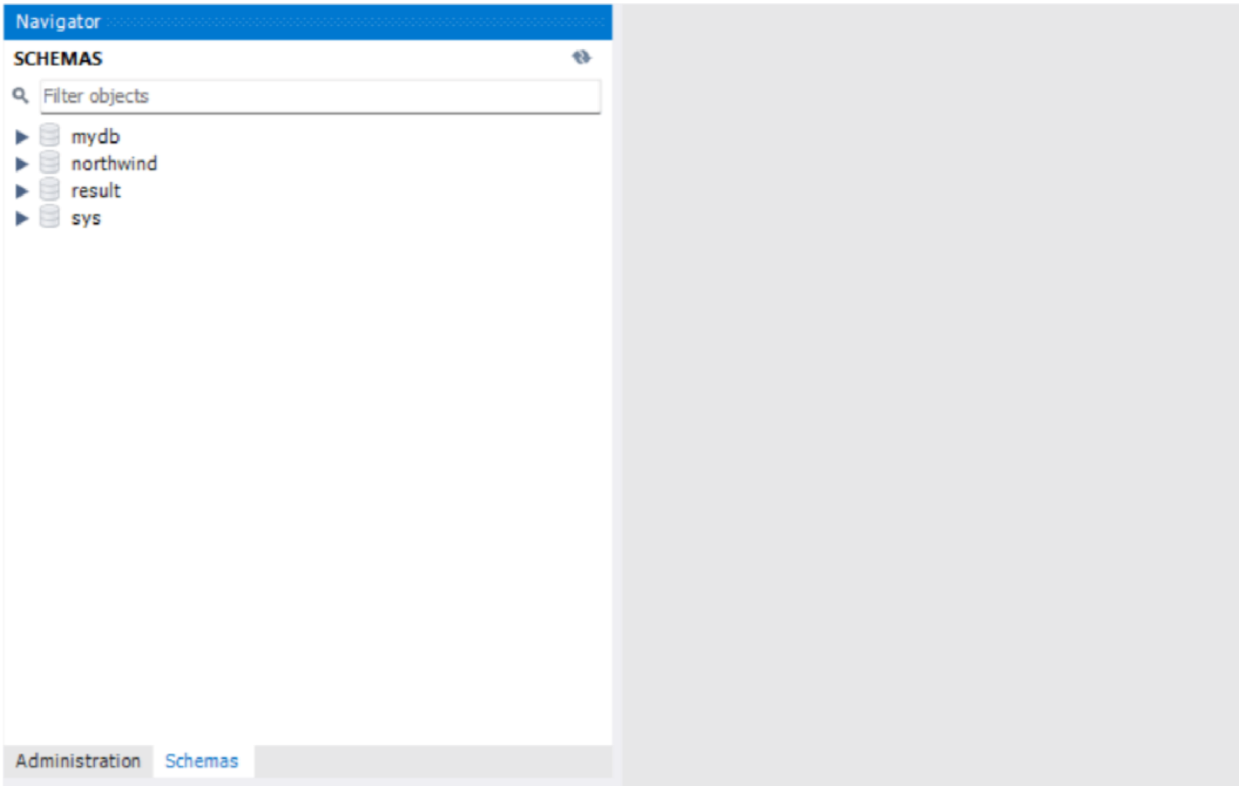
Step 1: Create a New Database

1. Open MySQL Workbench:

- Launch MySQL Workbench and connect to your MySQL server (e.g., Local Instance MySQL).
- Enter your root password if prompted.

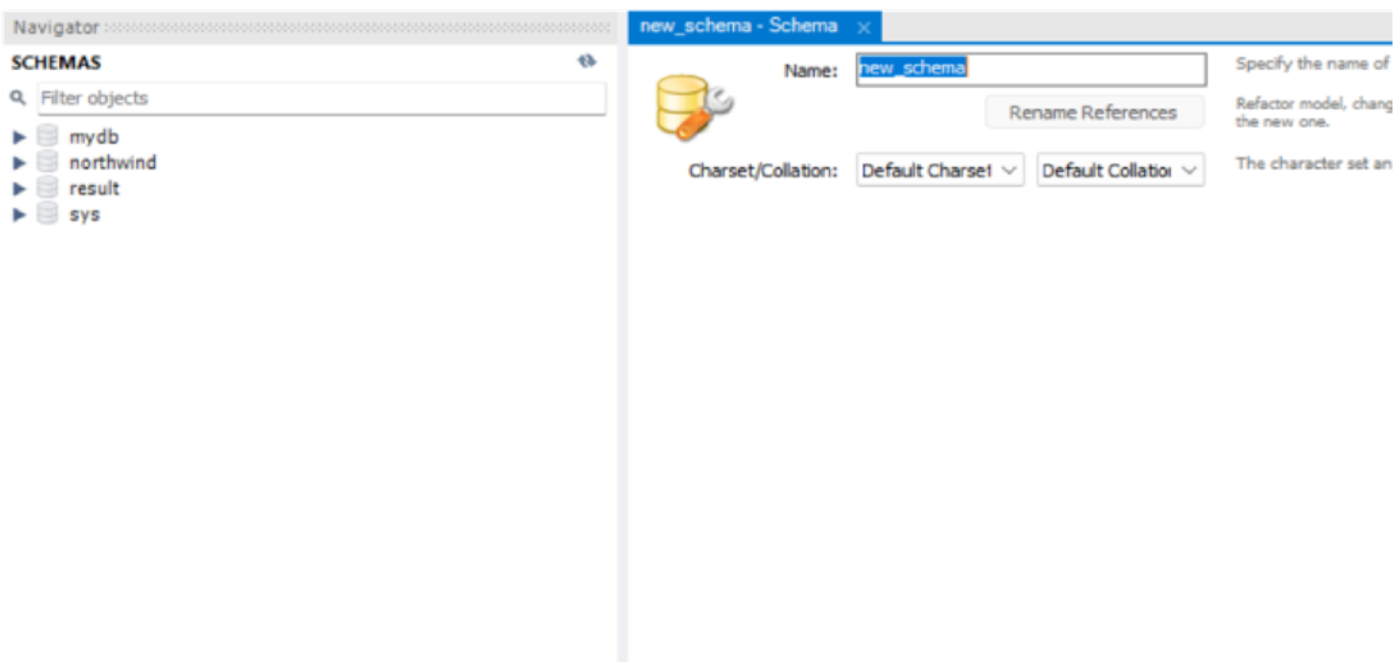
2. Go to the "Schemas" Pane:

- Locate the Schemas section on the left-hand side.



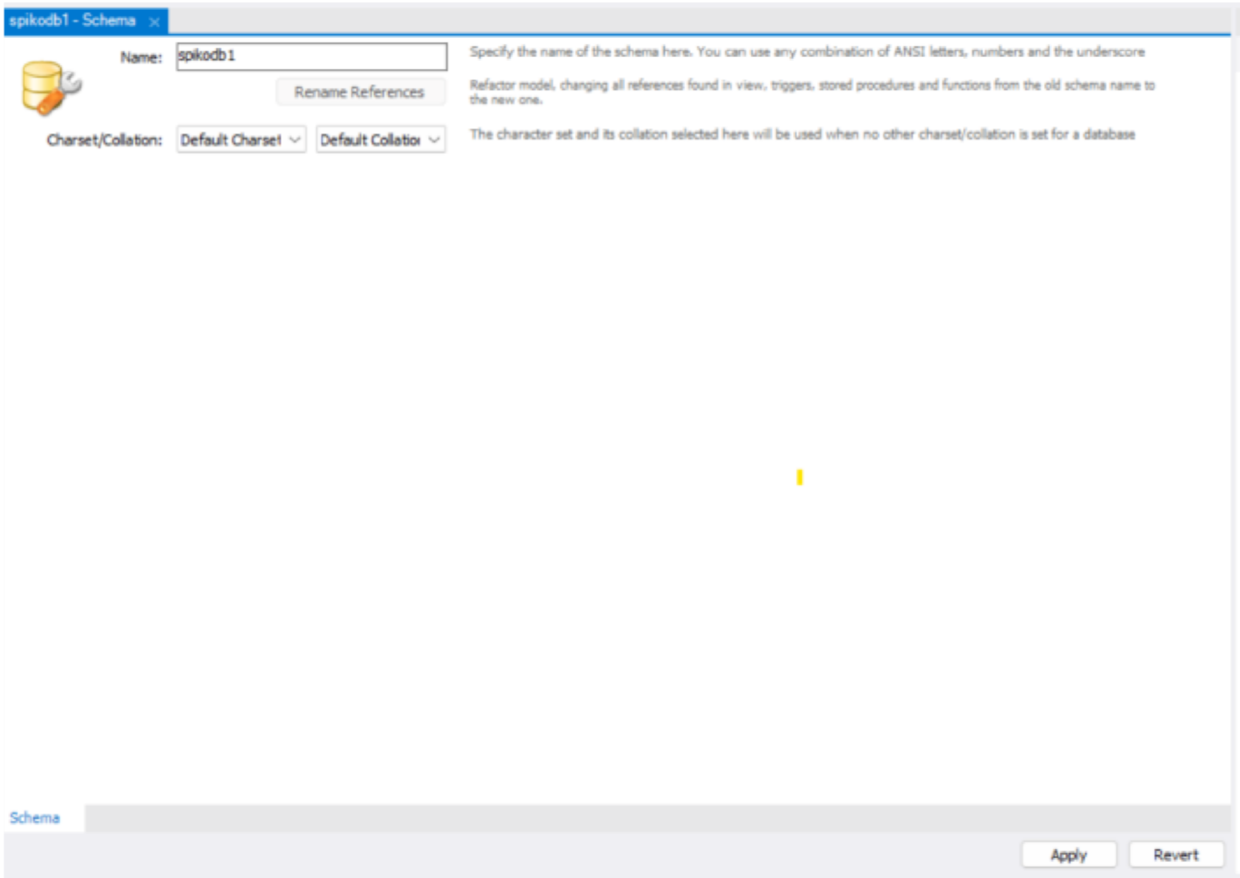
3. Create a New Database:

- Right-click anywhere in the Schemas pane and select "Create Schema...".
- A new window titled "Create Schema" will appear.



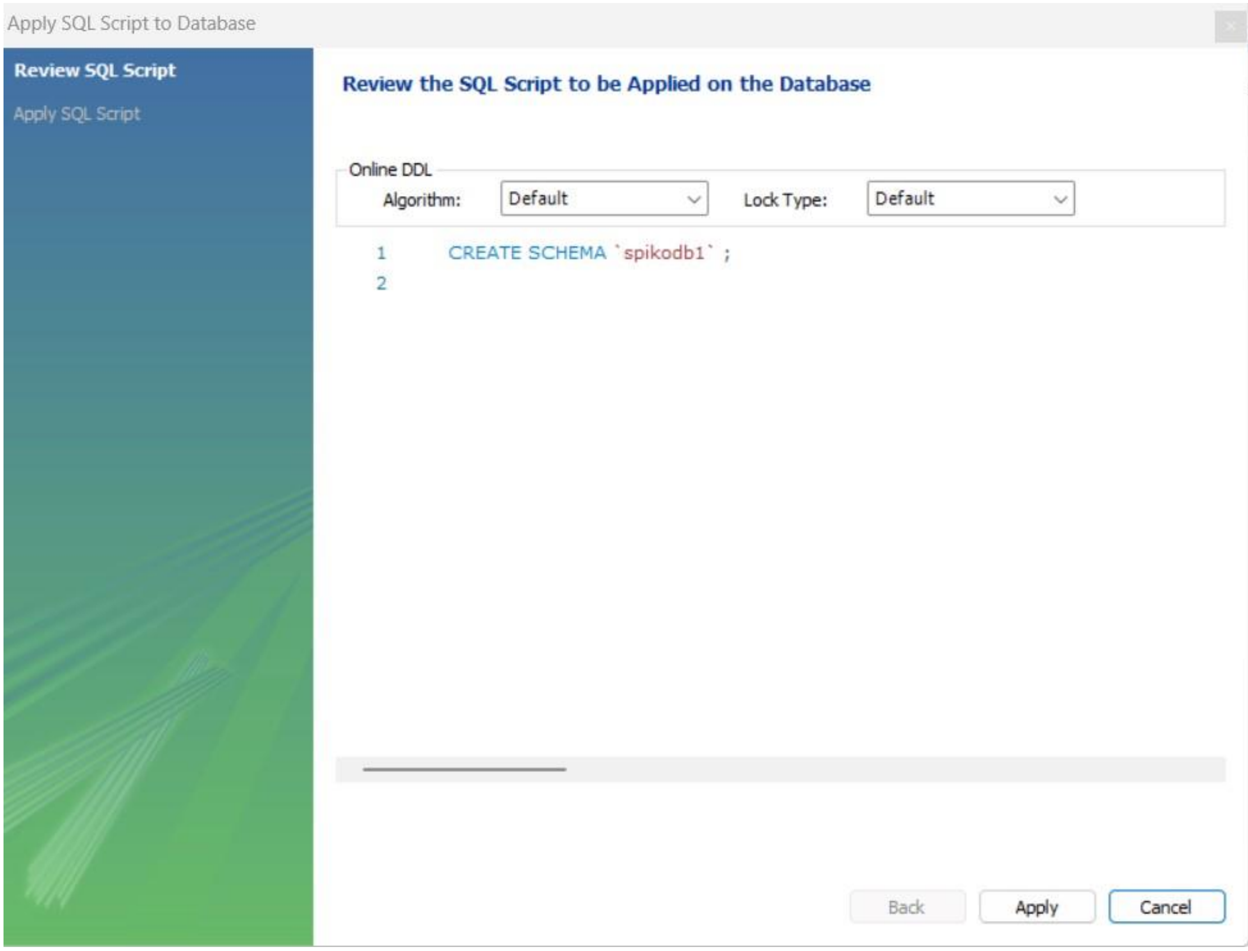
4. Provide the Database Name:

- In the Name field, enter the name of the database (e.g., spikodb1).
- Leave other settings (Character Set and Collation) as default.



5. Click "Apply":

- Review the automatically generated SQL script (e.g., CREATE DATABASE spikodb1;).
- Click Apply to execute the script.



6. Confirm Database Creation:

- Check for the success message and click Finish.
- Refresh the Schemas pane to see the new database (spikodb1).

Step 2: Create a Table

1. Set the Default Schema:

- Right-click on the newly created database (spikodb1) in the Schemas pane and select "Set as Default Schema".

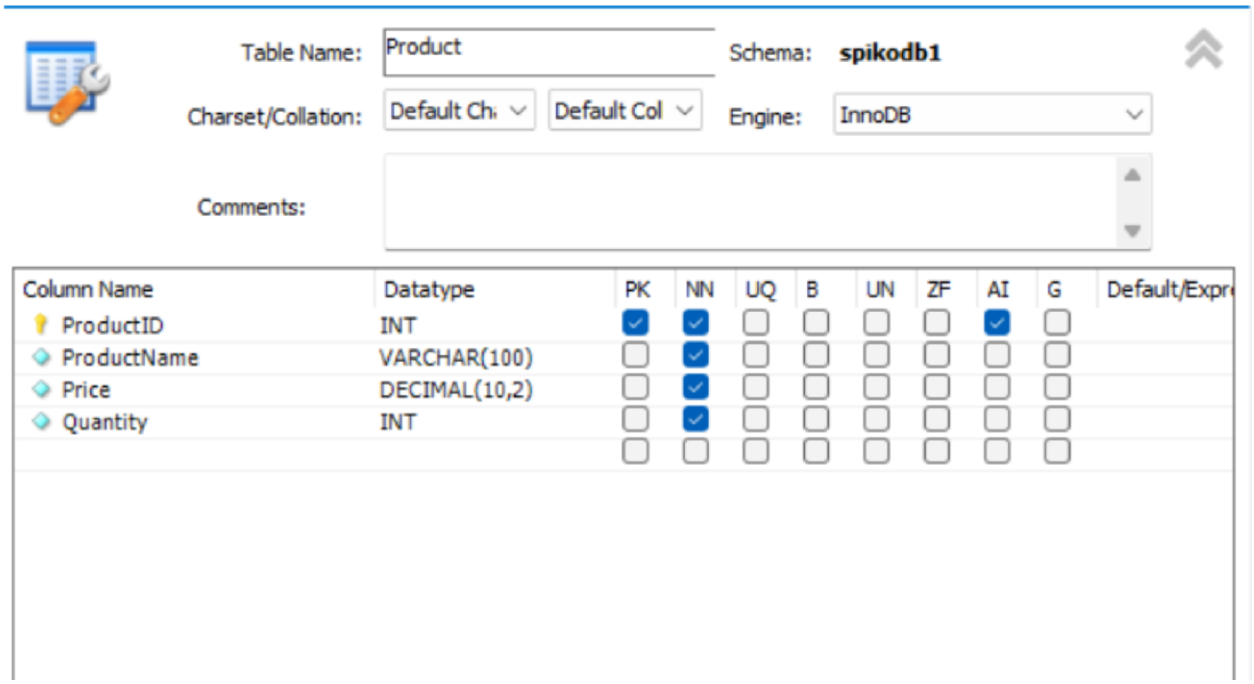
2. Open the Table Creation Wizard:

- Expand the spikodb1 database in the Schemas pane.
- Right-click on the Tables folder and select "Create Table...".

3. Define the Table Structure:

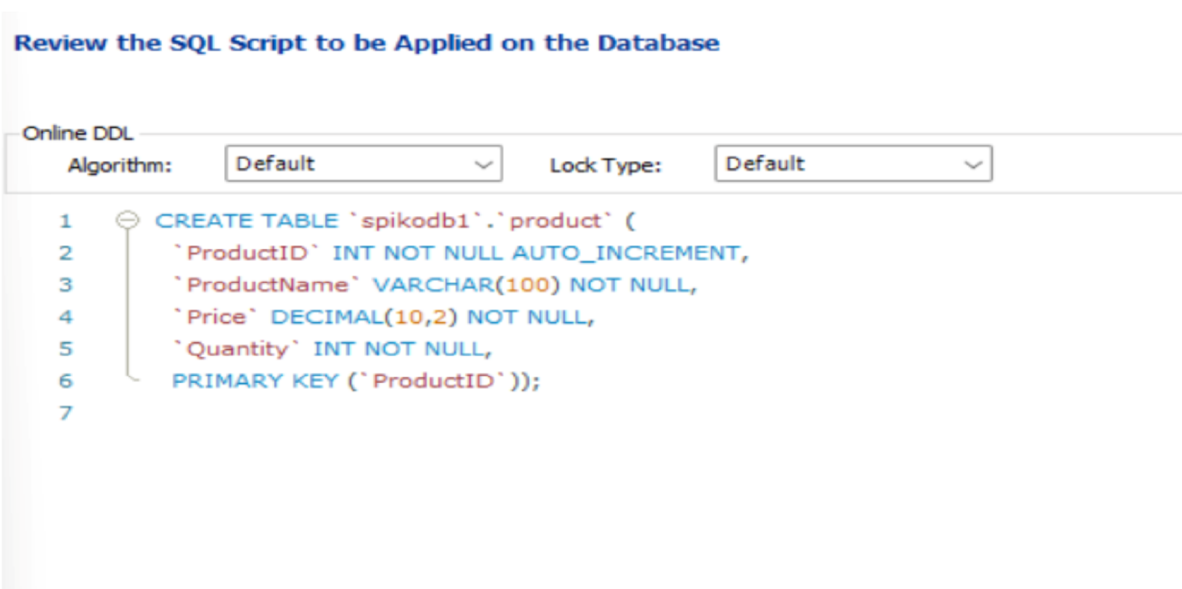
- Table Name: Enter Product in the Table Name field.
- Add columns by filling out the grid: ▪ Column Name: ProductID
  ▪ Data Type: INT

▪ Check PK (Primary Key), NN (Not Null), and AI (Auto Increment).
▪ Column Name: ProductName ▪ Data Type: VARCHAR(100) ▪ Check NN (Not Null).
▪ Column Name: Price
▪ Data Type: DECIMAL(10,2) ▪ Check NN (Not Null).
▪ Column Name: Quantity ▪ Data Type: INT
▪ Check NN (Not Null).



4. Click "Apply":

- Review the SQL script (e.g., CREATE TABLEProduct...).
- Click Apply to execute the script.



5. Confirm Table Creation:

- Check for the success message and click Finish.
- Expand the Tables folder in spikodb1 to see the Product table.

Step 3: Insert Data into the Table

1. Open the Table Editor:

- In the Schemas pane, expand the Tables folder under spikodb1.
- Right-click on the Product table and select "Select Rows - Limit 1000".

2. Switch to Edit Mode:

- In the result grid, click the Edit (pencil icon) at the bottom-right corner.

3. Enter Data:

- Add rows directly into the table by filling out the grid:

ProductName: Laptop, Price: 1000.00,

Quantity: 10

ProductName: Smartphone, Price: 500.00,

Quantity: 20

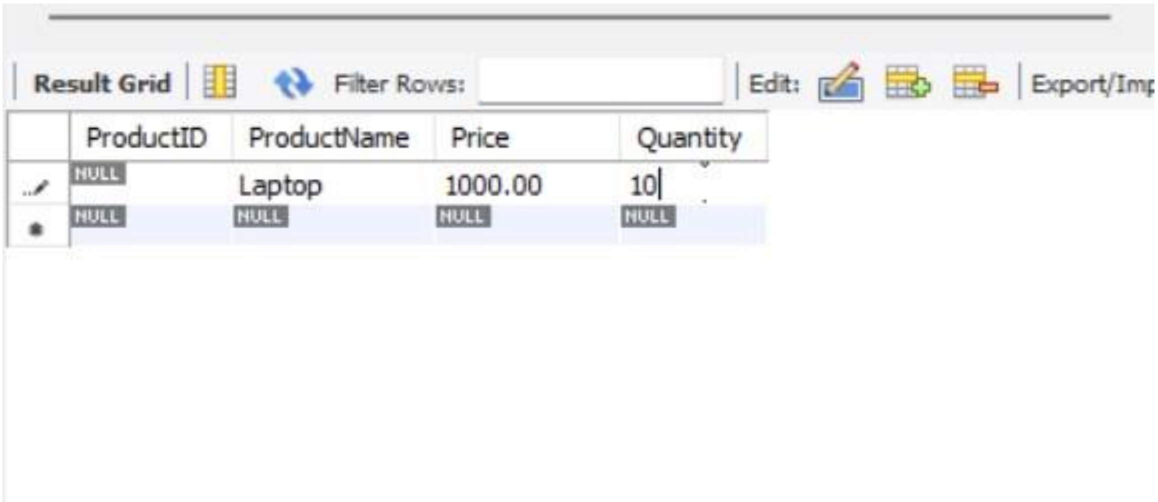
ProductName: Headphones, Price: 50.00,

Quantity: 50

ProductName: Keyboard, Price: 25.00,

Quantity: 30

ProductName: Mouse, Price: 15.00, Quantity:40



4.Apply Changes:

- Click Apply to save the data into the database.
- Confirm the SQL INSERT statements in the wizard and click Apply.

5. Verify the Data:

- Refresh the data grid to ensure the rows are successfully added.

Step 4: Verify the Table and Data

1. Check Table Data:

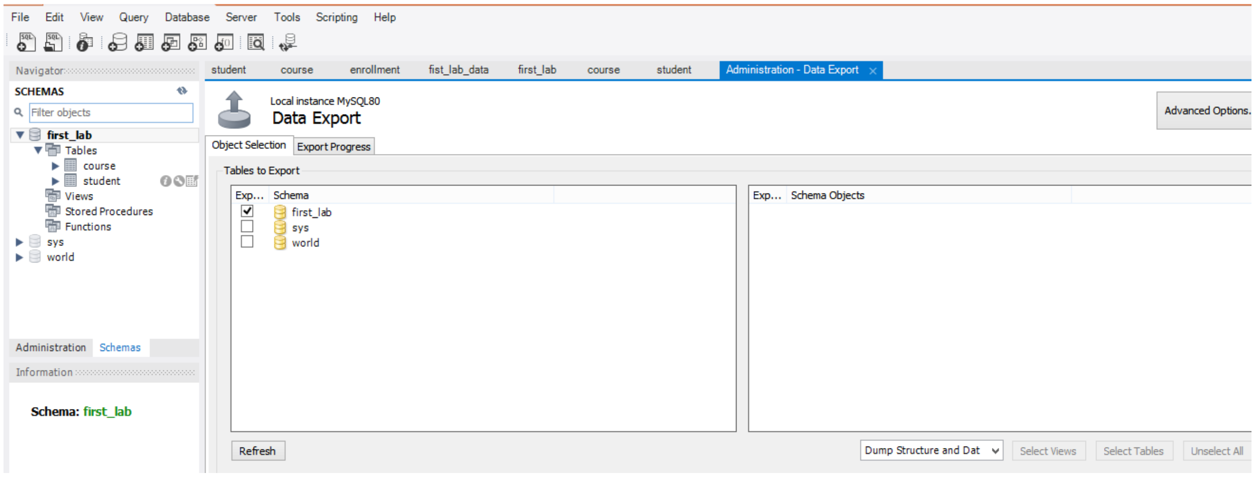
- Right-click on the Product table and select "Select Rows - Limit 1000" to view the inserted data.

2. Output Example:

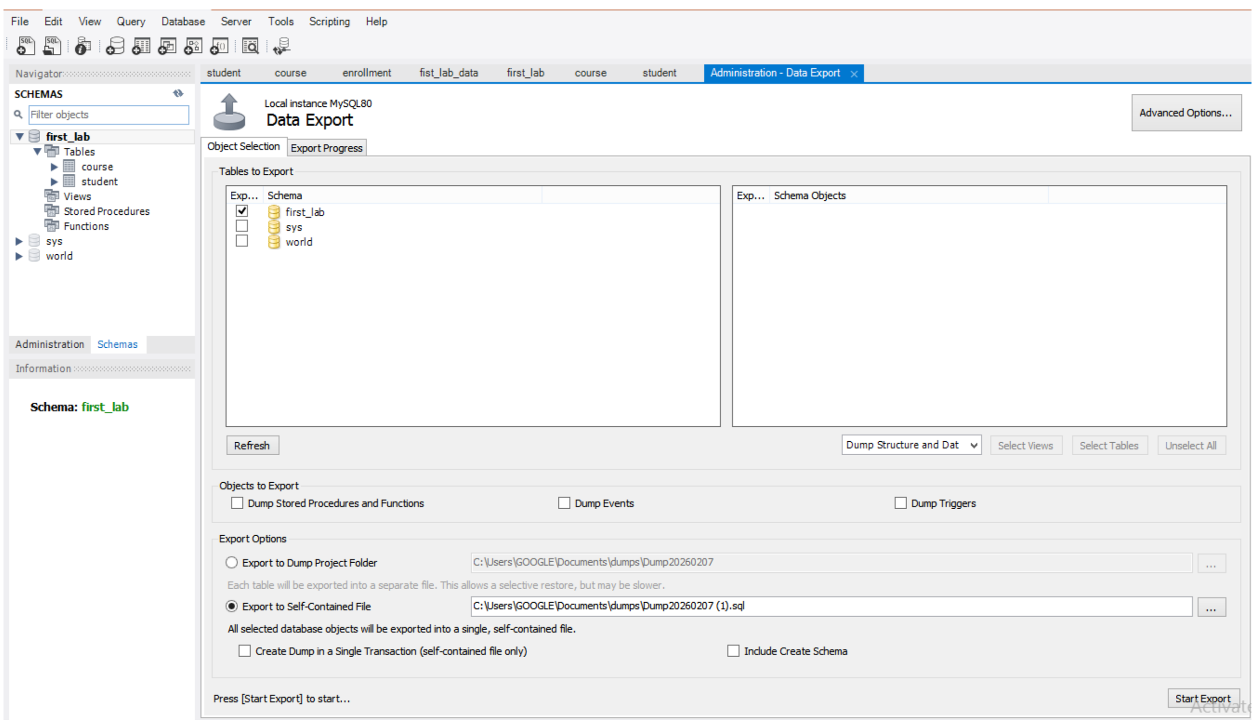
ProductID	ProductName	Price	Quantity
1	Laptop	1000.00	10
2	Smartphone	500.00	20
3	Headphones	50.00	50
4	Keyboard	25.00	30
5	Mouse	15.00	40

Lab Task 6: Export Database

1. In the left-hand menu, click “Server” → “Data Export”.
2. You will see a list of all databases on your server.
3. Tick the checkbox next to the database you want to export.

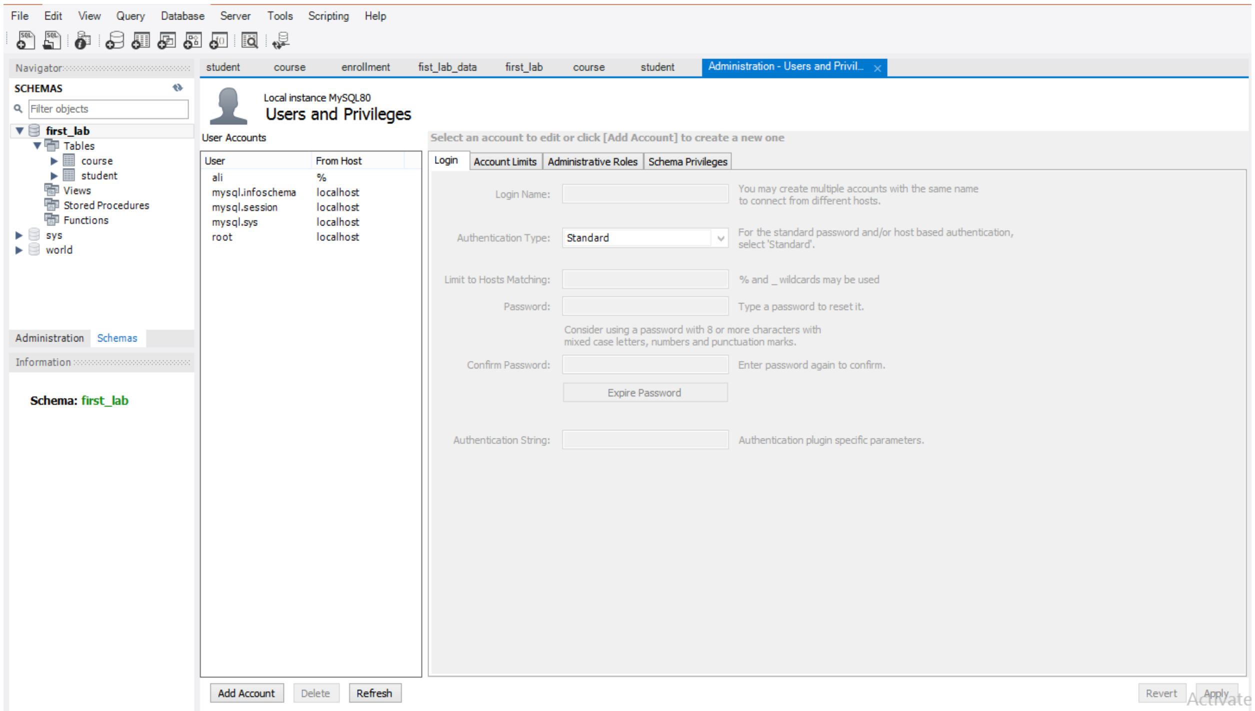


4. Export to Self-Contained File → saves everything (tables, data, schema) in one .sql file.
5. Make sure "Dump Structure and Data" is selected if you want both schema and data.
6. Click the "Export to Self-Contained File" box and choose where to save your .sql file.
7. Give it a name and click Start Export at the bottom-right.

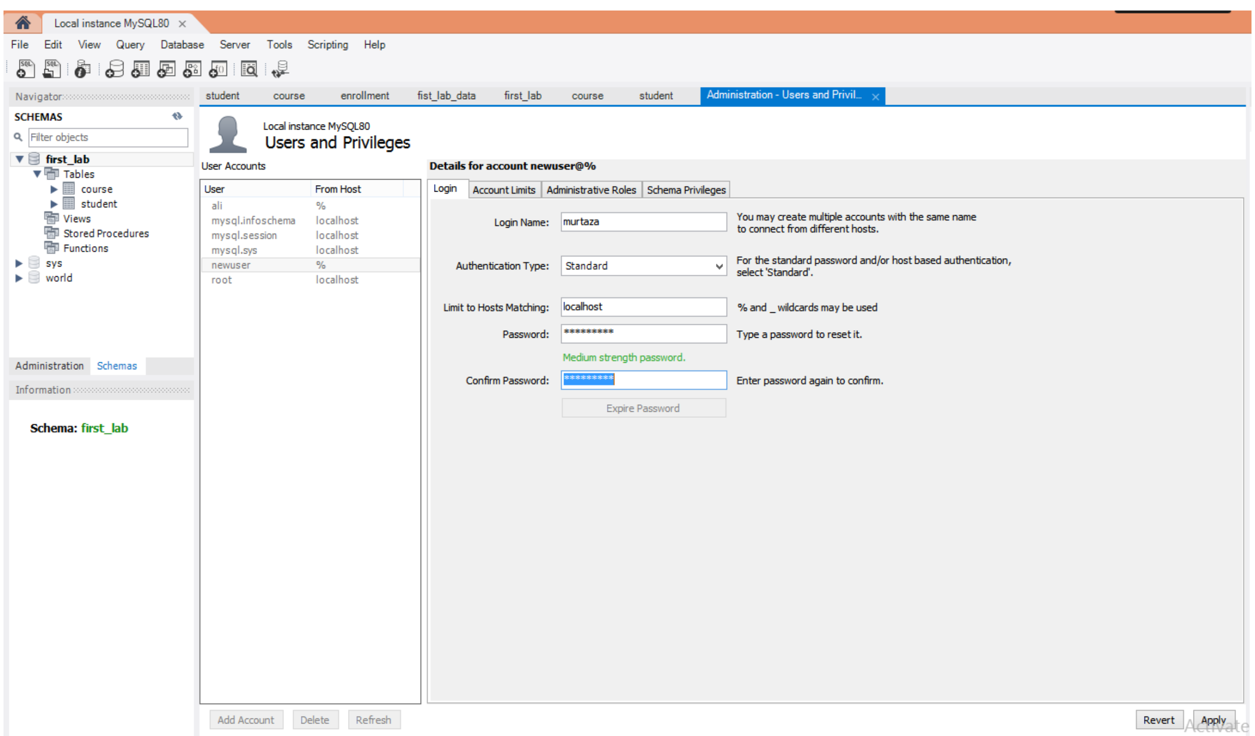


Lab Task 6: Create User+Permissions

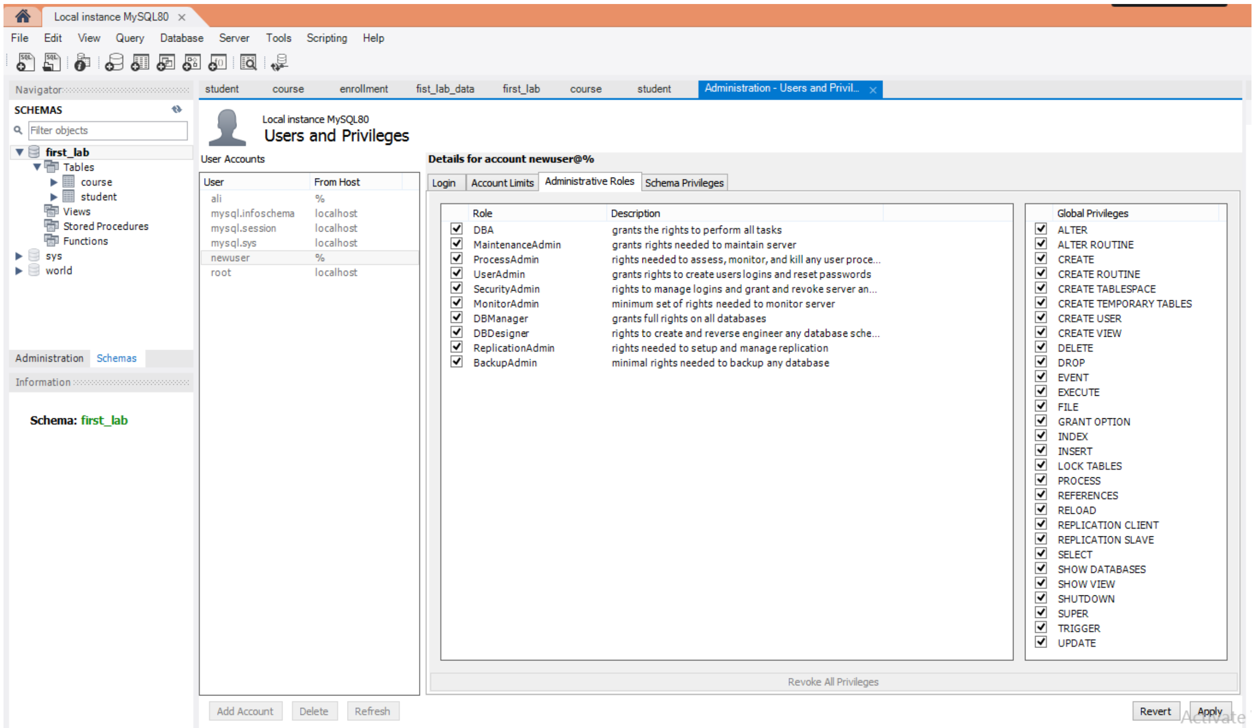
1. From the top menu, click "Server" and then "Users and Privileges".



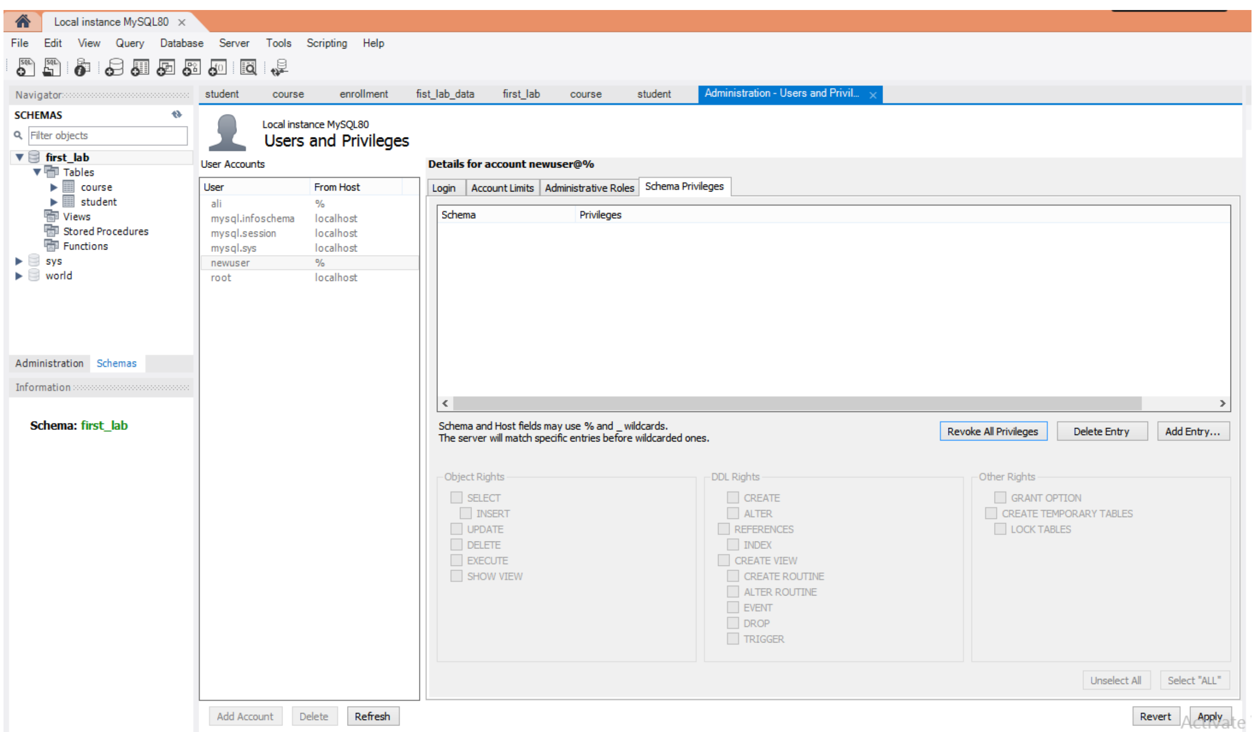
2. Click “Add Account” and fill in the details.



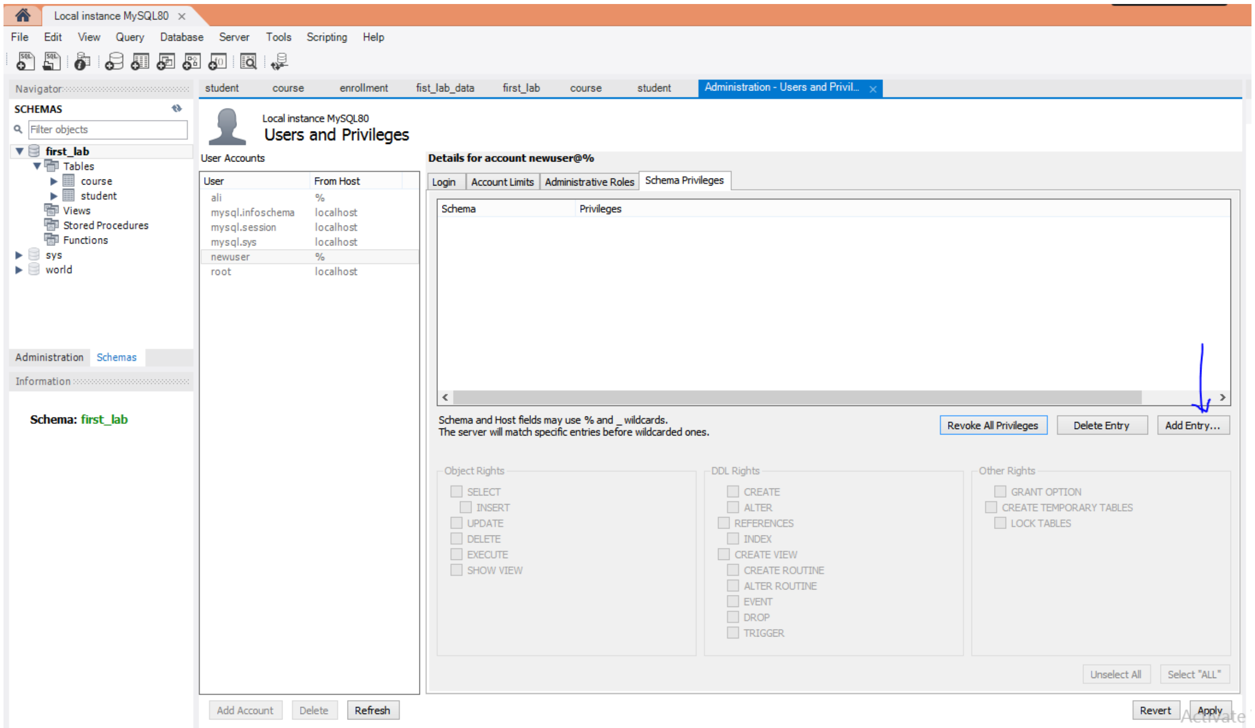
3. Go to the “Administrative Roles” tab to give predefined roles.
   - For full access: check DBA
   - For limited access: skip DBA and set individual privileges below.



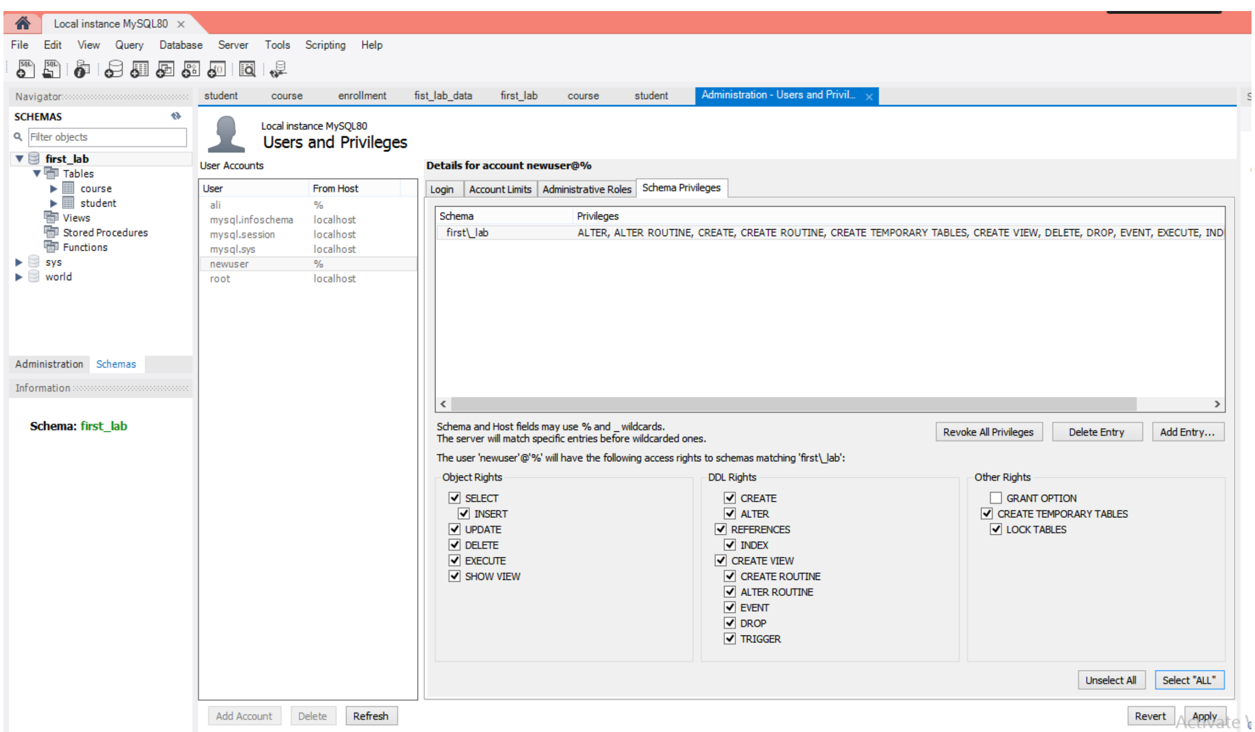
4. Or go to “Schema Privileges” tab to assign database-specific privileges.



5. Click “Add Entry”.



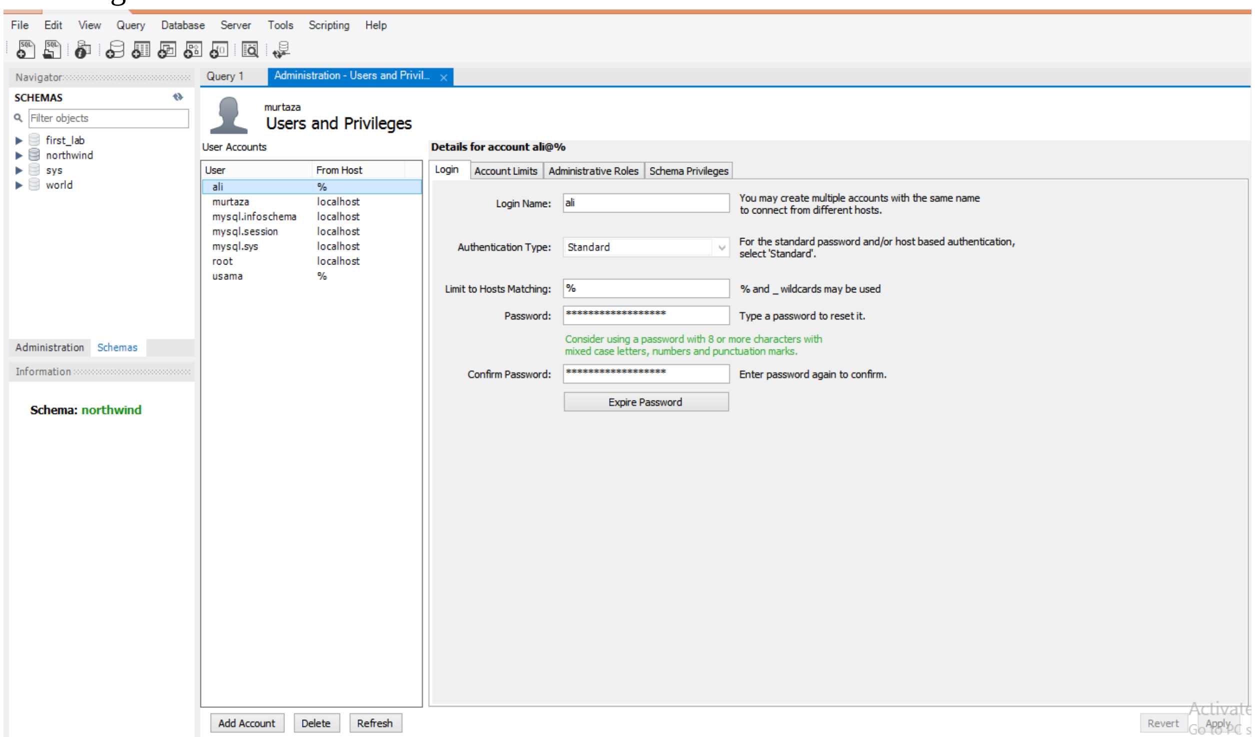
6. Choose the database and check the privileges you want.



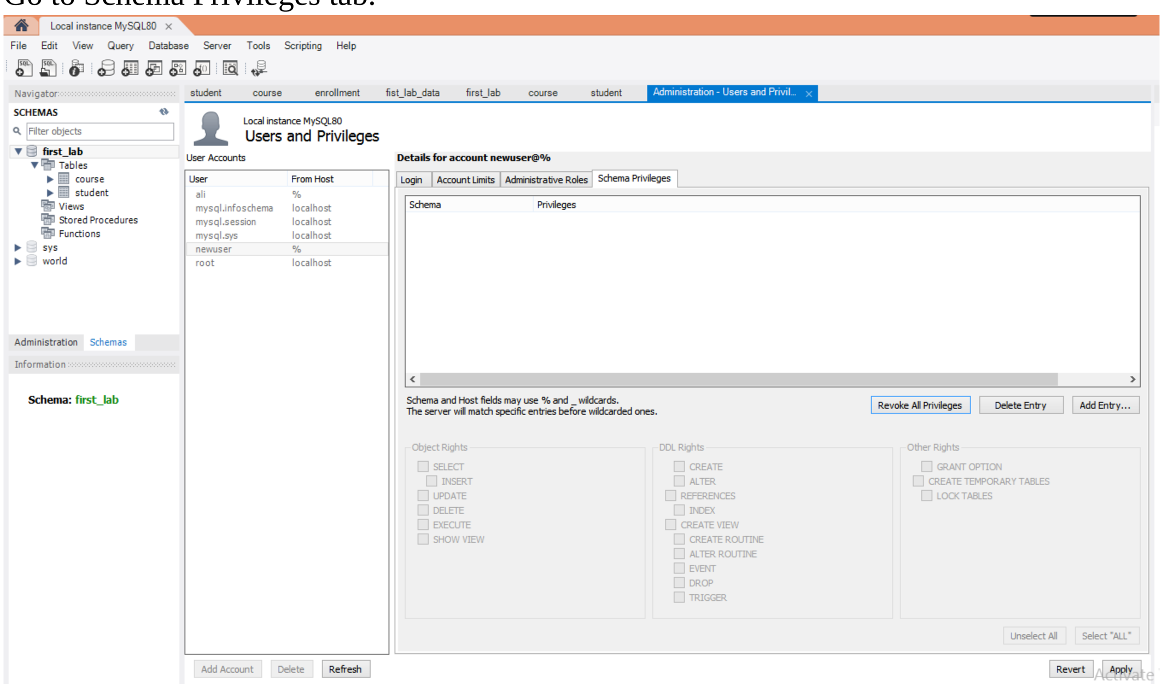
7. Click “Apply”.
8. Close the admin session and open a new connection in Workbench to test connection.

Lab Task7: Connect from another PC

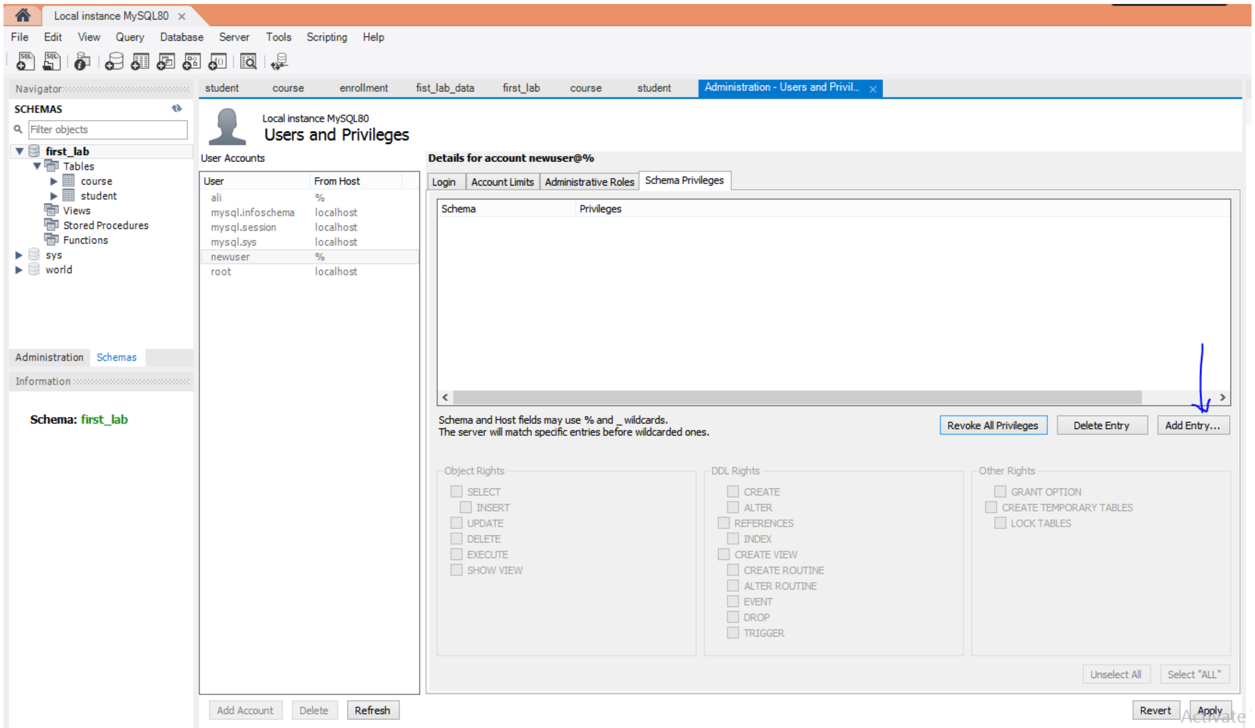
1. Open MySQL Workbench.
2. Go to Server and then Users and Privileges.
3. Click Add Account and fill in the details and make Limit Connectivity to Hosts Matching: %



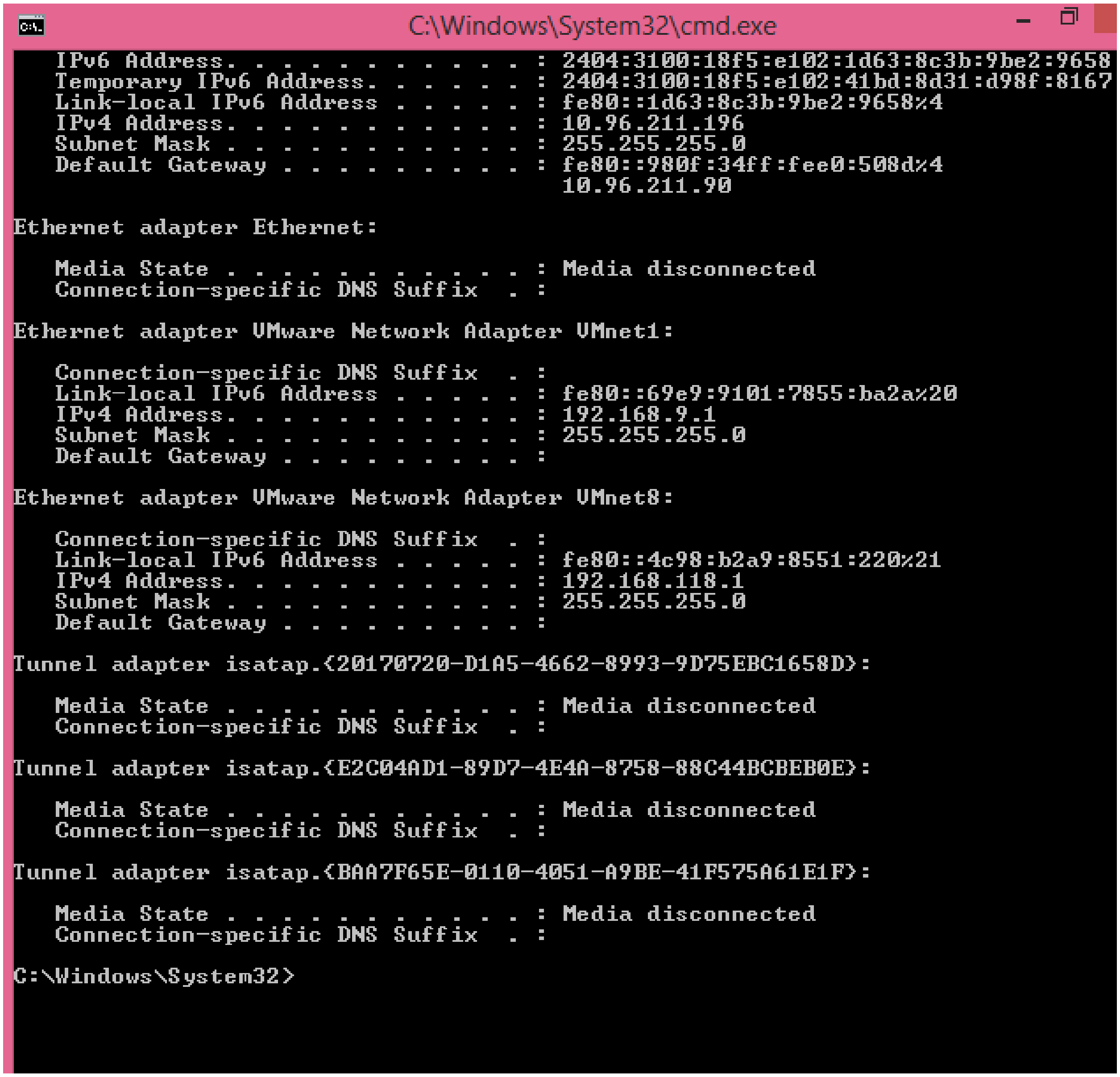
4. Go to Schema Privileges tab.



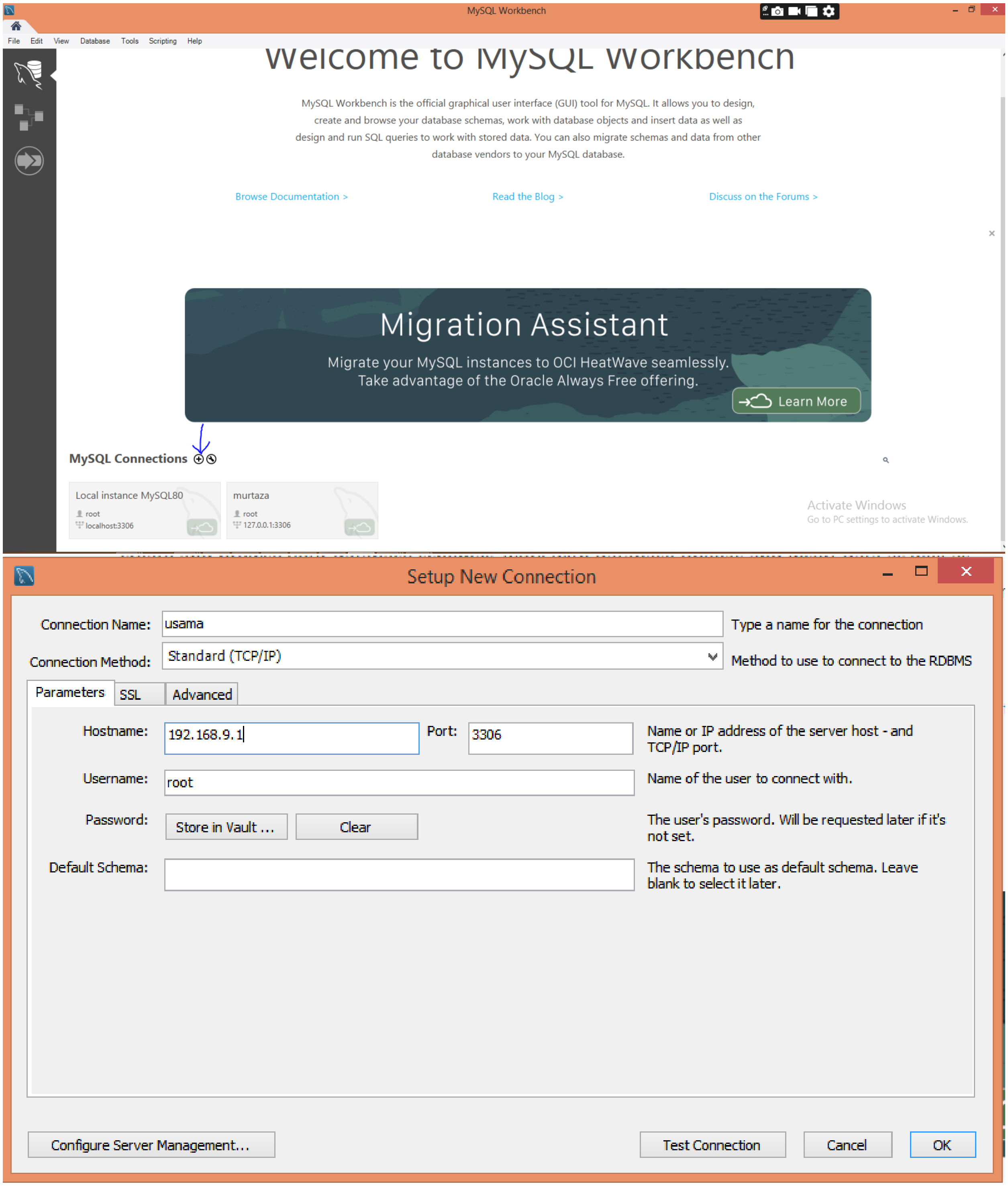
5. Click Add Entry.



6. Select database and check all privileges and then click apply.
7. Open Command Prompt on your PC/laptop and run ipconfig on cmd and note the IPv4 Address.



8. Open MySQL Workbench on the client PC.
9. Click + to create a new connection and enter the details and add IPv4 Address of server PC as hostname and test the connection.



10. Save and open the connection → now you can access first_lab remotely.

Home Tasks

Part 1:

1. Create a new database named TestDB_2025_CS_X

2. Create a new table named student using 5 to 6 attributes

3. Generate Scripts of your database

4. Restore your schema to another machine

5. Generate scripts of data

6. Restore data to other system as well

Part 2:

Task 1: Connect to the first_lab Database

- Objective: Ensure the database connection is set up properly.
- Steps:

1. Open MySQL Workbench.

2. Connect to your MySQL server by selecting your connection (e.g., Local Instance MySQL).

3. Verify that the first_lab database is visible in the Schemaspane.

Task 2: List All Tables

- Objective: Identify the tables in the first_lab database.
- Steps:

1. Expand the first_lab schema in the Schemas pane.

2. Note down the names of all tables (e.g., Customers, Orders, Products, etc.).

3. Output Example:

Tables in first_lab:

- Customers
- orders
- products
- employees
- categories

Task 3: View Table Structures

Objective: Understand the columns and data types in key tables

Steps:

1. Right-click on a table (e.g., Customers) in the Schemas pane.

2. Select "Table Inspector" or "View Table Details".

3. Note the column names, data types, and constraints (e.g., primary keys).

Task 4: Record Key Details

- Objective: Document the structure of a few key tables.
- Example for Customers:

Column Name	Data Type	Key/Constraint
CustomerID	INT	Primary Key
CustomerName	VARCHAR(45)	Not Null
ContactName	VARCHAR(45)	–
Country	VARCHAR(45)	_

Task 5: Browse Data

- **Objective: View a sample of the data stored in each table.**
- **Steps:**

1. Right-click on a table (e.g., Customers) in the Schemas pane.

2. Select "Select Rows - Limit 1000" to view the data.

3. Review the first few rows of data to understand the type of information stored.

Task 6: Document Examples

- Objective: Record a few rows of data from key tables for reference.
- Example for Customers:

CustomerID	CustomerName	ContactName	Country
1	John	9876543210	Pakistan

Task 7: Prepare a Lab Report

- Objective: Summarize findings in a concise report.

- Content to Include:

1. List of Tables: Document all tables in the database.
2. Table Structures: Provide column details for 2-3 key tables.
3. Sample Data: Include a 3 rows of data from each table.